



HTTP/3

HTTP/3 is the third major version of the Hypertext Transfer Protocol used to exchange information on the World Wide Web, complementing the widely-deployed HTTP/1.1 and HTTP/2. Unlike previous versions which relied on the well-established TCP (published in 1974),^[2] HTTP/3 uses QUIC (officially introduced in 2021),^[3] a multiplexed transport protocol built on UDP.^[4]

HTTP/3 uses similar semantics compared to earlier revisions of the protocol, including the same request methods, status codes, and message fields, but encodes them and maintains session state differently. However, partially due to the protocol's adoption of QUIC, HTTP/3 has lower latency and loads more quickly in real-world usage when compared with previous versions: in some cases over four times as fast than with HTTP/1.1 (which, for many websites, is the only HTTP version deployed).^{[5][6]}

As of September 2024, HTTP/3 is supported by more than 95% of major web browsers in use^[7] and 34% of the top 10 million websites.^[8] It has been supported by Chromium (and derived projects including Google Chrome, Microsoft Edge, Samsung Internet, and Opera)^[9] since April 2020 and by Mozilla Firefox since May 2021.^{[7][10]} Safari 14 implemented the protocol but it remains disabled by default.^[11]

History

HTTP/3 originates from an Internet Draft adopted by the QUIC working group. The original proposal was named "HTTP/2 Semantics Using The QUIC Transport Protocol",^[12] and later renamed "Hypertext Transfer Protocol (HTTP) over QUIC".^[13]

On 28 October 2018 in a mailing list discussion, Mark Nottingham, Chair of the IETF HTTP and QUIC Working Groups, proposed renaming HTTP-over-QUIC to HTTP/3, to "clearly identify it as another binding of HTTP semantics to the wire protocol [...] so people understand its separation from QUIC".^[14] Nottingham's proposal was accepted by fellow IETF members a few days later. The HTTP working group was chartered to assist the QUIC working group during the design of HTTP/3, then assume responsibility for maintenance after publication.^[15]

Support for HTTP/3 was added to Chrome (Canary build) in September 2019 and then eventually reached stable builds, but was disabled by a feature flag. It was enabled by default in April 2020.^[9] Firefox added support for HTTP/3 in November 2019 through a feature flag^{[7][16][17]} and started enabling it by default in April 2021 in Firefox 88.^{[7][10]} Experimental support for HTTP/3 was added to Safari Technology Preview on April 8, 2020^[18] and was included with Safari 14 that ships with iOS 14 and macOS 11,^{[11][19]} but it's still disabled by default as of Safari 16, on both macOS and iOS.

On 6 June 2022, IETF published HTTP/3 as a Proposed Standard in RFC 9114 (<https://datatracker.ietf.org/doc/html/rfc9114>).^[1]

HTTP/3

International standard	<u>RFC 9114</u> (https://datatracker.ietf.org/doc/html/rfc9114) ^[1] (HTTP/3 also uses the completed <u>QUIC</u> protocol described in RFC 9000 and related RFCs such as RFC 9001)
Developed by	<u>IETF</u>
Introduced	June 2022
Website	https://httpwg.org/specs/rfc9114.html

Internet history timeline

Early research and development:

- 1960–1964: RAND networking concepts developed
- 1962–1964: ARPA networking ideas
- 1965: NPL network concepts conceived
- 1966: Merit Network founded
- 1967: ARPANET planning begins
- 1967: Symposium on Operating Systems Principles
- 1969: NPL followed by the ARPANET carry their first packets
- 1970: Network Information Center (NIC)
- 1971: Tymnet switched-circuit network
- 1972: Merit Network's packet-switched network operational
- 1972: Internet Assigned Numbers Authority (IANA) established
- 1973: CYCLADES network demonstrated

Comparison with HTTP/1.1 and HTTP/2

HTTP semantics are consistent across versions: the same request methods, status codes, and message fields are typically applicable to all versions. The differences are in the mapping of these semantics to underlying transports. Both HTTP/1.1 and HTTP/2 use TCP as their transport. HTTP/3 uses QUIC, a transport layer network protocol which uses user space congestion control over the User Datagram Protocol (UDP). The switch to QUIC aims to fix a major problem of HTTP/2 called "head-of-line blocking": because the parallel nature of HTTP/2's multiplexing is not visible to TCP's loss recovery mechanisms, a lost or reordered packet causes all active transactions to experience a stall regardless of whether that transaction was impacted by the lost packet. Because QUIC provides native multiplexing, lost packets only impact the streams where data has been lost.

Proposed DNS resource records SVCB (service binding) and HTTPS would allow connecting without first receiving the Alt-Svc header via previous HTTP versions, therefore removing the 1 RTT of handshaking of TCP.^{[20][21]} There is client support for HTTPS resource records since Firefox 92, iOS 14, reported Safari 14 support, and Chromium supports it behind a flag.^{[22][23][24]}

Implementations

Client

Browser support for HTTP/3

Browser	Version implemented (disabled by default)		Version shipped (enabled by default)		Comment
Chrome	Stable build (79)	December 2019	87 ^[7]	April 2020 ^[25]	Earlier versions implemented other drafts of QUIC
Edge	Stable build (79)	December 2019	87	April 2020	Edge 79 was the first version based on Chromium
Firefox	Stable build (72.0.1)	January 2020	88 ^[10]	April 2021 ^[26]	
Safari	Stable build (14.0)	September 2020	16.4	March 2023	Apple is testing HTTP/3 support on some Safari users starting with Safari 16.4. ^[27]

Libraries

Open-source libraries that implement client or server logic for QUIC and HTTP/3 include^[28]

- 1973: PARC Universal Packet development begins
- 1974: Transmission Control Program specification published
- 1975: Telenet commercial packet-switched network
- 1976: X.25 protocol approved and deployed on public data networks
- 1978: Minitel introduced
- 1979: Internet Activities Board (IAB)
- 1980: USENET news using UUCP
- 1980: Ethernet standard introduced
- 1981: BITNET established

Merging the networks and creating the Internet:

- 1981: Computer Science Network (CSNET)
- 1982: TCP/IP protocol suite formalized
- 1982: Simple Mail Transfer Protocol (SMTP)
- 1983: Domain Name System (DNS)
- 1983: MILNET split off from ARPANET
- 1984: OSI Reference Model released
- 1985: First .COM domain name registered
- 1986: NSFNET with 56 kbit/s links
- 1986: Internet Engineering Task Force (IETF)
- 1987: UUNET founded
- 1988: NSFNET upgraded to 1.5 Mbit/s (T1)

- 1988: Morris worm
- 1988: Complete Internet protocol suite
- 1989: Border Gateway Protocol (BGP)
- 1989: PSINet founded, allows commercial traffic
- 1989: Federal Internet Exchanges (FIX East|FIXes)
- 1990: GOSIP (without TCP/IP)
- 1990: ARPANET decommissioned
- 1990: Advanced Network and Services (ANS)
- 1990: UUNET/Altnet allows commercial traffic
- 1990: Archie search engine
- 1991: Wide area information server (WAIS)
- 1991: Gopher
- 1991: Commercial Internet eXchange (CIX)
- 1991: ANS CO+RE allows commercial traffic
- 1991: World Wide Web (WWW)
- 1992: NSFNET upgraded to 45 Mbit/s (T3)
- 1992: Internet Society (ISOC) established
- 1993: Classless Inter-Domain Routing (CIDR)
- 1993: InterNIC established
- 1993: AOL added USENET access
- 1993: Mosaic web browser released
- 1994: Full text web search engines

- 1994: North American Network Operators' Group (NANOG) established

Commercialization, privatization, broader access leads to the modern Internet:

- 1995: New Internet architecture with commercial ISPs connected at NAPs
- 1995: NSFNET decommissioned
- 1995: GOSIP updated to allow TCP/IP
- 1995: very high-speed Backbone Network Service (vBNS)
- 1995: IPv6 proposed
- 1996: AOL changes pricing model from hourly to monthly
- 1998: Internet Corporation for Assigned Names and Numbers (ICANN)
- 1999: IEEE 802.11b wireless networking
- 1999: Internet2/Abilene Network
- 1999: vBNS+ allows broader access
- 2000: Dot-com bubble bursts
- 2001: New top-level domain names activated
- 2001: Code Red I, Code Red II, and Nimda worms
- 2003: UN World Summit on the Information Society (WSIS) phase I
- 2003: National LambdaRail founded
- 2004: UN Working Group on Internet Governance (WGIG)

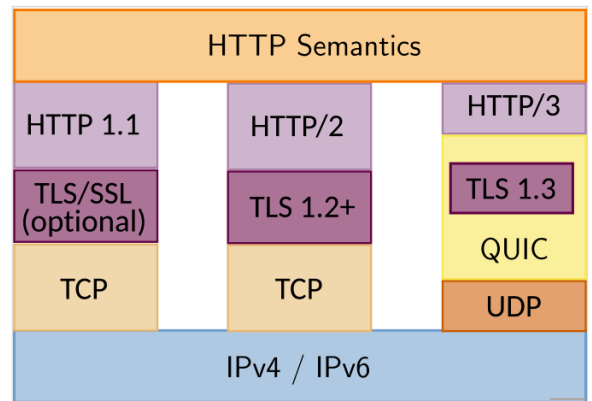
- 2005: UN WSIS phase II
- 2006: First meeting of the Internet Governance Forum
- 2010: First internationalized country code top-level domains registered
- 2012: ICANN begins accepting applications for new generic top-level domain names
- 2013: Montevideo Statement on the Future of Internet Cooperation
- 2014: NetMundial international Internet governance proposal
- 2016: ICANN contract with U.S. Dept. of Commerce ends, IANA oversight passes to the global Internet community on October 1st

Examples of Internet services:

- 1989: AOL dial-up service provider, email, instant messaging, and web browser
- 1990: IMDb Internet movie database
- 1994: Yahoo! web directory
- 1995: Amazon online retailer
- 1995: eBay online auction and shopping
- 1995: Craigslist classified advertisements
- 1995: AltaVista search engine
- 1996: Outlook (formerly Hotmail) free web-based e-mail
- 1996: RankDex search engine

- 1997: Google Search
- 1997: Babel Fish automatic translation
- 1998: Yahoo Groups (formerly Yahoo! Clubs)
- 1998: PayPal Internet payment system
- 1998: Rotten Tomatoes review aggregator
- 1999: 2ch Anonymous textboard
- 1999: i-mode mobile internet service
- 1999: Napster peer-to-peer file sharing
- 2000: Baidu search engine
- 2001: 2chan Anonymous imageboard
- 2001: BitTorrent peer-to-peer file sharing
- 2001: Wikipedia, the free encyclopedia
- 2003: LinkedIn business networking
- 2003: Myspace social networking site
- 2003: Skype Internet voice calls
- 2003: iTunes Store
- 2003: 4chan Anonymous imageboard
- 2003: The Pirate Bay, torrent file host
- 2004: Facebook social networking site
- 2004: Podcast media file series
- 2004: Flickr image hosting
- 2005: YouTube video sharing
- 2005: Reddit link voting
- 2005: Google Earth virtual globe
- 2006: Twitter microblogging

- 2007: WikiLeaks
anonymous news and
information leaks
- 2007: Google Street
View
- 2007: Kindle, e-reader
and virtual bookshop
- 2008: Amazon Elastic
Compute Cloud (EC2)
- 2008: Dropbox cloud-
based file hosting
- 2008: Encyclopedia of
Life, a collaborative
encyclopedia intended
to document all living
species
- 2008: Spotify, a DRM-
based music streaming
service
- 2009: Bing search
engine
- 2009: Google Docs,
Web-based word
processor,
spreadsheet,
presentation, form, and
data storage service
- 2009: Kickstarter, a
threshold pledge
system
- 2009: Bitcoin, a digital
currency
- 2010: Instagram, photo
sharing and social
networking
- 2011: Google+, social
networking
- 2011: Snapchat, photo
sharing
- 2012: Coursera,
massive open online
courses
- 2016: TikTok, video
sharing and social
networking



Protocol stack of HTTP/3 compared to HTTP/1.1 and HTTP/2

Libraries implementing HTTP/3

Name	Client	Server	Programming language	Company	Repository
lsquic	Yes	Yes	C	LiteSpeed	https://github.com/litespeedtech/lsquic
nghttp3	Yes	Yes	C		https://github.com/nghttp2/nghttp3
h2o	No	Yes	C		https://github.com/h2o/h2o
libcurl ^{[29][30]}	Yes	No	C		https://github.com/curl/curl
MsQuic ^[31]	Yes	Yes	C	Microsoft	https://github.com/microsoft/msquic
proxygen	Yes	Yes	C++	Facebook	https://github.com/facebook/proxygen#quic-and-http3
Cronet	Yes	Yes	C++	Google	https://github.com/chromium/chromium/tree/main/net/quic
.NET ^[32]	Yes	Yes	C# (using MsQuic) ^[33]	Microsoft	https://github.com/dotnet
quic-go	Yes	Yes	Go		https://github.com/quic-go/quic-go
http3	Yes	Yes	Haskell		https://github.com/kazu-yamamoto/http3
Kwik	Yes	Yes	Java		https://github.com/ptrd/kwik
Flupke	Yes	Yes	Java		https://bitbucket.org/pjtr/flupke
aioquic	Yes	Yes	Python		https://github.com/aiortc/aioquic
quiche	Yes	Yes	Rust	Cloudflare	https://github.com/cloudflare/quiche
neqo	Yes	Yes	Rust	Mozilla	https://github.com/mozilla/neqo
quinn	Yes	Yes	Rust		https://github.com/quinn-rs/quinn
s2n-quic	Yes	Yes	Rust	Amazon Web Services	https://github.com/aws/s2n-quic

Server

- On 7 June 2021, [LiteSpeed Web Server](#) (and [OpenLiteSpeed](#)) 6.0.2 was released and became the first version to enable HTTP/3 by default.^[34]
- [Caddy](#) web server v2.6.0 (released 20 September 2022) has HTTP/3 enabled by default.^[35]
- [Nginx](#) supports HTTP/3 since 1.25.0 (released 23 May 2023). A technology preview of nginx with HTTP/3 support was released in June 2020.^[36] Binary packages of nginx with HTTP/3 support have been released in February 2023.^[37]
- [Cloudflare](#) distributes a patch for nginx that integrates the quiche HTTP/3 library into it.^[38]

- Microsoft IIS support for HTTP/3 is enabled natively with Windows Server 2022/Windows 11.^[39]
- HAProxy supports HTTP/3 over QUIC since version 2.6 released on 31 May 2022.^{[40][41]}
- Nimble Streamer supports HTTP/3 since 4.1.8–1^[42] for HTTP-based protocols.

See also



- Fast and Secure Protocol — Terminal command scheme used to transfer data

References

1. M. Bishop, ed. (June 2022). *HTTP/3* (<https://datatracker.ietf.org/doc/html/rfc9114>). Internet Engineering Task Force. doi:10.17487/RFC9114 (<https://doi.org/10.17487%2FRFC9114>). ISSN 2070–1721 (<https://search.worldcat.org/issn/2070-1721>). RFC 9114 (<https://datatracker.ietf.org/doc/html/rfc9114>). *Proposed Standard*.
2. V. Cerf; Y. Dalal; C. Sunshine (December 1974). *SPECIFICATION OF INTERNET TRANSMISSION CONTROL PROGRAM* (<https://datatracker.ietf.org/doc/html/rfc675>). Network Working Group. doi:10.17487/RFC0675 (<https://doi.org/10.17487%2FRFC0675>). RFC 675 (<https://datatracker.ietf.org/doc/html/rfc675>). *Obsolete*. Obsoleted by RFC 7805 (<https://datatracker.ietf.org/doc/html/rfc7805>). NIC 2. INWG 72.
3. J. Iyengar; M. Thomson, eds. (May 2021). *QUIC: A UDP-Based Multiplexed and Secure Transport* (<https://datatracker.ietf.org/doc/html/rfc9000>). Internet Engineering Task Force. doi:10.17487/RFC9000 (<https://doi.org/10.17487%2FRFC9000>). ISSN 2070–1721 (<https://search.worldcat.org/issn/2070-1721>). RFC 9000 (<https://datatracker.ietf.org/doc/html/rfc9000>). *Proposed Standard*.
4. "What is HTTP/3?" (<https://www.cloudflare.com/learning/performance/what-is-http3/>). *Cloudflare*. Archived (<https://web.archive.org/web/20220704035921/https://www.cloudflare.com/learning/performance/what-is-http3/>) from the original on 4 July 2022. Retrieved 12 July 2022.
5. Perna, Gianluca; Trevisan, Martino; Giordano, Danilo; Drago, Idilio (1 April 2022). "A first look at HTTP/3 adoption and performance" (<https://www.sciencedirect.com/science/article/pii/S0140366422000421>). *Computer Communications*. **187**: 115–124. doi:10.1016/j.comcom.2022.02.005 (<https://doi.org/10.1016%2Fj.comcom.2022.02.005>). hdl:11368/3025202 (<https://hdl.handle.net/11368%2F3025202>). ISSN 0140–3664 (<https://search.worldcat.org/issn/0140-3664>). S2CID 246936473 (<https://api.semanticscholar.org/CorpusID:246936473>).
6. "HTTP/3 is Fast" (<https://requestmetrics.com/web-performance/http3-is-fast>). *Request Metrics*. Retrieved 1 July 2022.
7. " "HTTP/3" | Can I use... Support tables for HTML5, CSS3, etc" (<https://caniuse.com/http3>). *caniuse.com*. Retrieved 11 August 2024.
8. "Usage of HTTP/3 for websites" (<https://w3techs.com/technologies/details/ce-http3>). *World Wide Web Technology Surveys*. W3Techs. Retrieved 11 August 2024.
9. "Enabling QUIC in tip-of-tree" (https://groups.google.com/a/chromium.org/g/net-dev/c/5M9Z5mtvg_Y/m/iw9co1VrBQAJ). *groups.google.com*. Retrieved 8 April 2021.
10. Damjanovic, Dragana (16 April 2021). "QUIC and HTTP/3 Support now in Firefox Nightly and Beta" (<https://hacks.mozilla.org/2021/04/quic-and-http-3-support-now-in-firefox-nightly-and-beta>). *Mozilla Hacks — the Web developer blog*. Retrieved 17 April 2021.
11. "Safari 14 Release Notes" (<https://developer.apple.com/documentation/safari-release-notes/safari-14-release-notes>). *developer.apple.com*. Retrieved 4 December 2020.
12. Shade, Robbie (8 July 2016). *HTTP/2 Semantics Using The QUIC Transport Protocol* (<https://datatracker.ietf.org/doc/html/draft-shade-quic-http2-mapping>). IETF. I-D draft-shade-quic-http2-mapping.
13. Cimpanu, Catalin (12 November 2018). "HTTP-over-QUIC to be renamed HTTP/3" (<https://www.zdnet.com/article/http-over-quic-to-be-renamed-http3/>). *ZDNet*. Retrieved 12 November 2018.
14. Nottingham, Mark (28 October 2018). "Identifying our deliverables" (https://mailarchive.ietf.org/arch/msg/quic/RLRs4nB1lwFCZ_7k0iuz0ZBa35s). *IETF Mail Archive*.

15. "Hypertext Transfer Protocol Charter" (<https://datatracker.ietf.org/doc/charter-ietf-httpbis/08/>). *ietf.org*. Retrieved 2 September 2020.
16. Daniel, Stenberg. "Daniel Stenberg announces HTTP/3 support in Firefox Nightly" (<https://twitter.com/bagder/status/1191482712739196928>). *Twitter*. Retrieved 5 November 2019.
17. Cimpanu, Catalin (26 September 2019). "Cloudflare, Google Chrome, and Firefox add HTTP/3 support" (<https://www.zdnet.com/article/cloudflare-google-chrome-and-firefox-add-http3-support/>). *ZDNet*. Retrieved 27 September 2019.
18. "Release Notes for Safari Technology Preview 104" (<https://webkit.org/blog/10264/release-notes-for-safari-technology-preview-104/>). *webkit.org*. 8 April 2020. Retrieved 7 August 2020.
19. Ng, Gary (23 June 2020). "Apple's Safari Adds Support for HTTP3 in iOS 14 and macOS 11" (<https://www.iphoneincanada.ca/news/apple-safari-http3-ios-14/>). *iphoneincanada.ca*. Retrieved 25 June 2021.
20. "HTTPS RR" (https://developer.mozilla.org/en-US/docs/Glossary/https_rr). *MDN*. Mozilla. Retrieved 25 October 2022.
21. Schwartz, Benjamin M.; Bishop, Mike; Nygren, Erik (12 June 2020). *Service binding and parameter specification via the DNS* (<https://datatracker.ietf.org/doc/html/draft-ietf-dnsop-svcb-https>). *IETF*. I-D draft-ietf-dnsop-svcb-https.
22. "Firefox 92 for developers" (<https://developer.mozilla.org/en-US/docs/Mozilla/Firefox/Releases/92>). Mozilla Corporation. 7 September 2021. Retrieved 25 October 2022.
23. "Feature: HTTP->HTTPS redirect for HTTPS DNS records" (<https://chromestatus.com/feature/5485544526053376>). Google Inc. Retrieved 25 October 2022.
24. Patrick Mevzek (24 August 2021). "What's the use case of SVCB (type 65, service binding) RR" (<https://serverfault.com/questions/1075522/whats-the-use-case-of-svcb-type-65-service-binding-rr>). Stack Exchange Inc. Retrieved 25 October 2022.
25. "Enabling QUIC in tip-of-tree" (https://groups.google.com/a/chromium.org/g/net-dev/c/5M9Z5mtvg_Y/m/iw9co1VrBQAJ). *groups.google.com*. Retrieved 9 April 2021.
26. "Firefox Release Owners – MozillaWiki" (https://wiki.mozilla.org/Release_Management/Release_owners). *wiki.mozilla.org*. Retrieved 9 April 2021.
27. Jen Simmons (4 April 2023). "HTTP/3 support shipped in Safari 14.0" (<https://github.com/Fyrd/caniuse/pull/6664#issuecomment-1496439786>). *GitHub*. Retrieved 7 April 2023.
28. "QUIC Implementations" (<https://github.com/quicwg/base-drafts/wiki/Implementations>). *GitHub*. Retrieved 8 April 2021.
29. "First HTTP/3 with curl" (<https://daniel.haxx.se/blog/2019/08/05/first-http-3-with-curl/>). Daniel Stenberg. 5 August 2019. Retrieved 2 October 2019.
30. "HTTP3 (and QUIC)" (<https://github.com/curl/curl/blob/master/docs/HTTP3.md>). Daniel Stenberg. 23 August 2023. Retrieved 27 August 2023.
31. "MsQuic is Open Source" (<https://techcommunity.microsoft.com/t5/networking-blog/msquic-is-open-source/ba-p/1345441>). 28 April 2020. Retrieved 28 April 2020.
32. "HTTP/3 support in .NET 6" (<https://devblogs.microsoft.com/dotnet/http-3-support-in-dotnet-6/>). 17 September 2021. Retrieved 17 September 2021.
33. "HTTP/3 support in .NET 6" (<https://devblogs.microsoft.com/dotnet/http-3-support-in-dotnet-6/>). *.NET Blog*. 17 September 2021. Retrieved 12 January 2022.
34. "LiteSpeed Web Server Release Log – LiteSpeed Technologies" (<https://www.litespeedtech.com/product/s/litespeed-web-server/release-log>). *www.litespeedtech.com*. Retrieved 12 February 2022. "Enable HTTP/3 v1 by default."
35. "Release 2.6.0 · caddyserver/caddy" (<https://github.com/caddyserver/caddy/releases/tag/v2.6.0>). *GitHub*. 22 September 2022. Retrieved 20 September 2022.
36. "Introducing a Technology Preview of NGINX Support for QUIC and HTTP/3" (<https://www.nginx.com/blog/introducing-technology-preview-nginx-support-for-quic-http-3/>). *NGINX*. 10 June 2020. Retrieved 11 June 2020.

37. "Binary Packages Now Available for the Preview NGINX QUIC+HTTP/3 Implementation" (<https://www.nginx.com/blog/binary-packages-for-preview-nginx-quic-http3-implementation/>). *NGINX*. 8 February 2023. Retrieved 30 March 2023.
38. "Experiment with HTTP/3 using NGINX and quiche" (<https://blog.cloudflare.com/experiment-with-http-3-using-nginx-and-quiche/>). *The Cloudflare Blog*. 17 October 2019. Retrieved 9 November 2019.
39. Tratcher. "Use ASP.NET Core with HTTP/3 on IIS" (<https://docs.microsoft.com/en-us/aspnet/core/host-and-deploy/iis/http3>). *docs.microsoft.com*. Retrieved 29 April 2022.
40. "Announcing HAProxy 2.6" (<https://www.haproxy.com/blog/announcing-haproxy-2-6/#http-3-over-quic>). *HAProxy Blog*. 31 May 2022.
41. "QUIC Implementation in HAProxy" (https://www.youtube.com/watch?v=e2aQBva_HH0). *HAProxyConf video presentation*. 25 January 2023.
42. "HTTP/3 and QUIC support in Nimble Streamer" (<https://softvelum.com/2025/02/http3-quic-support-nimble-streamer/>). *NGINX*. 14 February 2025. Retrieved 17 February 2025.

External links

- [Official website](https://httpwg.org/specs/rfc9114.html) (<https://httpwg.org/specs/rfc9114.html>) 
- [IETF QUIC Working Group](https://github.com/quicwg) (<https://github.com/quicwg>) on [GitHub](#)
- [HTTP/3 explained](https://http3-explained.haxx.se/) (<https://http3-explained.haxx.se/>) ([Daniel Stenberg](#))
- [HTTP/3 on caniuse.com](https://caniuse.com/http3) (<https://caniuse.com/http3>)
- [List of QUIC implementations on the IETF QUIC Working Group Wiki](https://github.com/quicwg/base-drafts/wiki/Implementations) (<https://github.com/quicwg/base-drafts/wiki/Implementations>)

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